

On the structure of finitely generated subgroups of branch groups

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10 July 2025

Motivations

- ▶ What are branch groups, and why we would like to study them;
- ▶ Profinite topology and finitely generated subgroups;
- ▶ Consequences.

Branch groups

- ▶ Groups with a rich subnormal subgroup structure ($H = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G$);
- ▶ Contains example of groups with exotic properties:
 - ▶ Finitely generated infinite torsion groups,
 - ▶ Groups of intermediate growth,
 - ▶ Amenable but non-elementary amenable groups;
- ▶ Naturally appears in the classification of just infinite groups (infinite groups whose proper quotients are finite);
- ▶ Admits a nice action on a rooted tree (and hence are residually finite);
- ▶ Many branch groups are recursively presented (a weaker notion than finitely generated).

The profinite topology

Any group G can be endowed with the **profinite topology**: the topology generated by all left cosets of finite index normal subgroups.

- ▶ G is residually finite if and only if $\{1\}$ is closed (for finitely generated groups: if and only if G acts faithfully on a locally finite rooted tree);
- ▶ If G is residually finite, recursively presented and has an algorithm listing all finite index normal subgroups, then the word problem is solvable.

Finitely generated branch groups with the *congruence subgroup property* have an algorithm listing all finite index normal subgroups.

Profinite topology: beyond residually finite groups

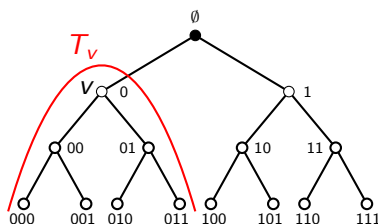
- ▶ A group G is **LERF** (locally extensively residually finite, or subgroup separable) if every finitely generated subgroup is closed in the profinite topology;
 - ▶ If G is LERF, recursively presented and has an algorithm listing all finite index normal subgroups, then the generalized word problem is solvable (subgroup membership for finitely generated subgroups),
 - ▶ Finite groups, f. g. abelian groups, virtually polycyclic groups, f. g. free groups, surface groups, limit groups, free metabelian groups, some right-angled Artin groups are LERF. Some branch groups are LERF.
- ▶ A group G has the **Ribes-Zalesskiĭ property** if for any n and every finitely generated subgroups $H_1, \dots, H_n$, the subset $H_1 H_2 \cdots H_n$ is closed;
 - ▶ $n = 0$ is residually finite, $n = 1$ is LERF,
 - ▶ Finite groups, f. g. abelian groups, f.g. free groups, limit groups, Kleinian groups and a few additional examples have the RZ property. Conjecture: some branch groups have the Ribes-Zalesskiĭ property.

Classification of finitely generated subgroups: other consequences

- ▶ Can be used to describe weakly maximal subgroups (subgroups maximal among infinite index subgroups);
- ▶ G has the **Howson property** if for H, K finitely generated subgroups, $H \cap K$ is still finitely generated.
 - ▶ Most finitely generated branch groups do not have the Howson property [Francoeur 2025+]. But one hope to have an algorithm to decide whenever $H \cap K$ is finitely generated or not.

Regular rooted trees

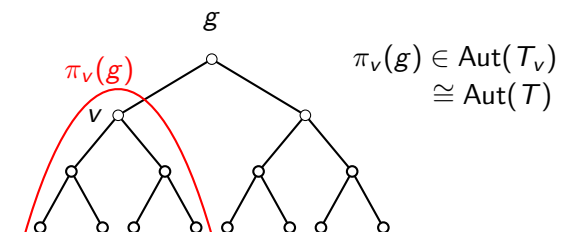
- ▶ $T = T_d$: the d -regular rooted tree (the root has degree d and each other vertex has degree $d + 1$);



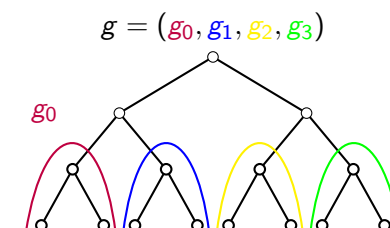
- ▶ Vertices of T_d are in bijection with finite words on the alphabet $\{0, \dots, d-1\}$ (root $\leftrightarrow \emptyset$ the empty word);
- ▶ The n^{th} level $\mathcal{L}_n$ of the tree is the set of vertices at distance n of the root;
- ▶ T_v is the subtree of T consisting of vertices below v .

Sections of elements of $\text{Aut}(T)$

- ▶ For v a vertex of T and $g \in \text{Stab}_{\text{Aut}(T)}(v)$, the **section** $\pi_v(g) = g|_v$ of g at v is the automorphism of T_v induced by g .



- ▶ Elements g that fixe $\mathcal{L}_n$ are usually described as the product of their sections:



Self-similar groups

Definition

A group $G \leq \text{Aut}(T)$ is **self-similar** if for every vertex v in T we have $\pi_v(\text{Stab}_G(v)) \leq G$.

Definition

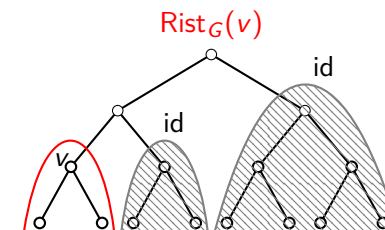
A group $G \leq \text{Aut}(T)$ is **self-replicating** (or fractal) if for every vertex v in T we have $\pi_v(\text{Stab}_G(v)) = G$.

Some subgroups of $\text{Aut}(T_d)$

Let $G \leq \text{Aut}(T_d)$. The following subgroups play an important role:

- ▶ Stabilizers of vertices $\text{Stab}_G(v)$;
- ▶ Pointwise stabilizers of levels $\text{Stab}_G(\mathcal{L}_n)$;
- ▶ **Rigid stabilizer** of vertices:

$$\begin{aligned} \text{Rist}_G(v) &:= \{g \in G \mid g \text{ acts trivially outside } T_v\} \\ &= \bigcap_{w \notin T_v} \text{Stab}_G(w) \end{aligned}$$



Some subgroups of $\text{Aut}(T_d)$

Let $G \leq \text{Aut}(T_d)$. The following subgroups play an important role:

- ▶ Stabilizers of vertices $\text{Stab}_G(v)$;
 - ▶ Pointwise stabilizers of levels $\text{Stab}_G(\mathcal{L}_n)$;
 - ▶ Rigid stabilizer of vertices $\text{Rist}_G(v)$,
 - ▶ **Rigid stabilizer of levels:** $\text{Rist}_G(\mathcal{L}_n) := \prod_{v \in \mathcal{L}_n} \text{Rist}_G(v)$.
- Carefull:** $\text{Rist}_G(\mathcal{L}_n) \neq \text{Rist}_{\text{Aut}(T)}(\mathcal{L}_n) \cap G$.

The congruence subgroup property

Definition

Let T be a rooted tree. A subgroup $G \leq \text{Aut}(T)$ has the **congruence subgroup property** if for any finite index subgroup H there exists n such that $\text{Stab}_G(n) \leq H$.

G has the congruence subgroup property if and only if the profinite topology (generated by the finite index normal subgroups and their cosets) and the *natural topology on $\text{Aut}(T)$* (generated by the $\text{Stab}_G(n)$ and their cosets) agree.

The congruence subgroup property provides an algorithm describing finite index normal subgroups of G . If G is recursively presented, then the word problem is solvable (and the generalized word problem also, provided that G is LERF).

Branch groups

Definition

A subgroup G of $\text{Aut}(T)$ is **branch** if for all n

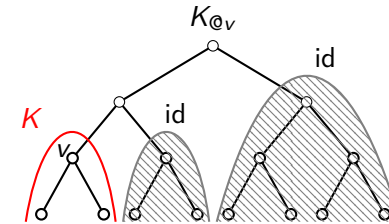
1. G acts transitively on $\mathcal{L}_n$,
2. $\text{Rist}_G(\mathcal{L}_n)$ is a finite index subgroup of G .

Regularly branch groups

Definition

A self-similar subgroup G of $\text{Aut}(T)$ is **regularly branch** if

1. G acts transitively on $\mathcal{L}_n$ for every n ;
2. There exists a finite index subgroup $K \leq G$ such that $K_{@v} \leq K$ for every vertex v , where $K_{@v} := \{g \in \text{Rist}(v) \mid \pi_v(g) \in K\}$.



Regularly branch groups are branch.

Regularly branch groups

Example

The first Grigorchuk group $\mathfrak{G}$, the Gupta-Sidki p -groups ($p \geq 3$ prime), torsion GGS groups (acting on T_p , $p \geq 3$ prime).

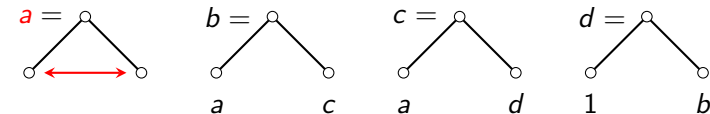
All these examples:

- ▶ Are regularly branch;
- ▶ Have the congruence subgroup property;
- ▶ Are recursively presented;
- ▶ Are infinite, just infinite, torsion, all their maximal subgroups are of finite index, ...

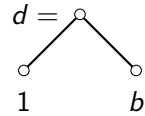
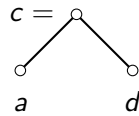
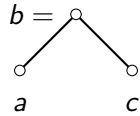
Moreover $\mathfrak{G}$ has intermediate growth, ...

The first Grigorchuk group

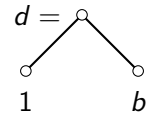
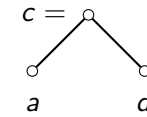
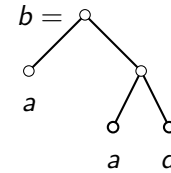
The first Grigorchuk group $\mathfrak{G} = \langle a, b, c, d \rangle$ acts on T_2 and is generated by



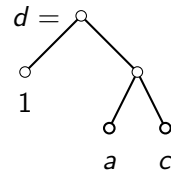
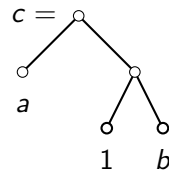
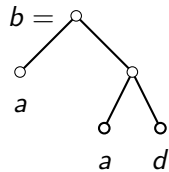
The first Grigorchuk group



The first Grigorchuk group

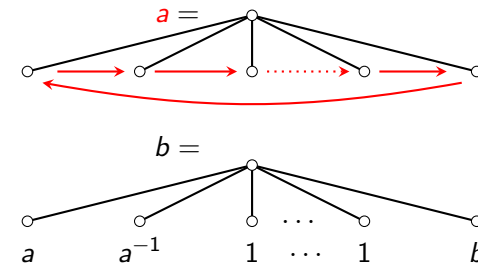


The first Grigorchuk group



The Gupta-Sidki p -group

The group G_p acts on T_p ($p \geq 3$ prime) and is generated by a and b , where



GGs groups

- ▶ Let $p \geq 3$ be a prime and let $\mathbf{e} = (e_0, \dots, e_{p-2})$ be a vector in $(\mathbf{Z}/p\mathbf{Z})^{p-1} \setminus \{0\}$. The GGS group $G_{\mathbf{e}} = \langle a, b \rangle$ with defining vector $\mathbf{e}$ is the subgroup of $\text{Aut}(T_p)$ generated by

$a = \text{cyclic permutation } (12 \dots p) \text{ of the first level vertices}$

$b = (a^{e_0}, \dots, a^{e_{p-2}}, b);$

- ▶ The group $G_{\mathbf{e}}$ is torsion if and only if $\sum_{i=0}^{p-2} e_i = 0$;
- ▶ The Gupta-Sidki p -group correspond to the special case $\mathbf{e} = (1, -1, 0, \dots, 0)$.

Some finitely generated subgroups

Let $G \leq \text{Aut}(T)$ be a finitely generated group regularly branch over K . Then the following are finitely generated subgroups of G :

- ▶ K ;
- ▶ $K_{\partial v}$ for every vertex v .

One can use the above to construct other finitely generated subgroups.

Diagonal subgroups

Let $G \leq \text{Aut}(T)$ be a finitely generated group regularly branch over K . Let $v_1, \dots, v_n$ be pairwise incomparable vertices and $\varphi_1, \dots, \varphi_n$ be automorphisms of K . This data define a **diagonal subgroup (over K)**:

$$\text{diag}(\varphi_1(K) \times \varphi_2(K) \times \varphi_3(K)) = \left\{ g = \begin{array}{c} \text{tree structure} \\ \varphi_1(k) \quad \varphi_2(k) \quad \varphi_3(k) \end{array} \mid k \in K \right\}$$

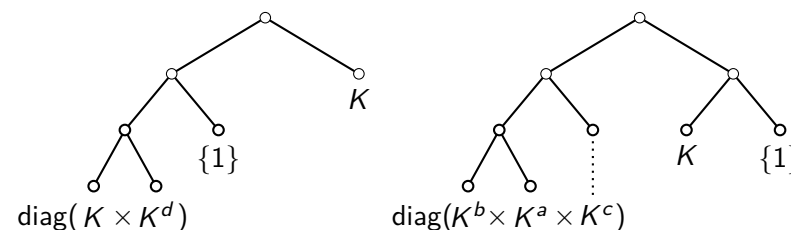
Diagonal subgroups are finitely generated.

Block subgroups

Definition

A **block subgroup (over K)** is a finite product of diagonal subgroups over K (such that all the corresponding vertices are incomparable).

Example



Block subgroups are finitely generated, as well as virtually block subgroups ($H \leq G$ such that there exists $B \leq H$ of finite index which is block).

Main result

Main theorem

Let G be either the first Grigorchuk group or a torsion GGS group (for some prime p). Then every finitely generated subgroup of G is virtually a block subgroup over K .

For the first Grigorchuk group, $K = \langle [a, b] \rangle^\emptyset \leq_{16} \mathfrak{S}$. For a GGS group $G = G_{\mathbf{e}}$, $K = \gamma_3(G) = \langle [G', G] \rangle \leq_{p^3} G$ if $\mathbf{e}$ is symmetric, or over $K = G' \leq_{p^2} G$ if $\mathbf{e}$ is not symmetric.

Abstract version

Let G be a regularly branch group and let RB be the set of subgroups on which G regularly branch. RB admits a unique maximal element, which is called **the maximal branching subgroup of G** .

Theorem

Let $G \leq \text{Aut}(T)$ be a finitely generated, self-replicating regularly branch group and let K be its maximal branching subgroup.

Suppose that:

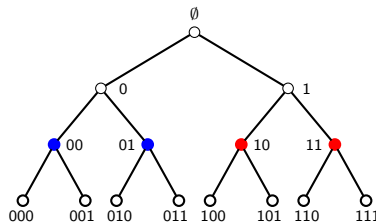
- ▶ G acts tree-primitively on T ;
- ▶ G has trivial branch kernel;
- ▶ G has the subgroup induction property.

Then every finitely generated subgroup of G is virtually a block subgroup.

Tree-primitive action

The action of a subgroup $G \leq \text{Aut}(T)$ on T needs to preserve the tree structure and therefore cannot be primitive, even when restricted to some level $\mathcal{L}_n$.

For example, if $T = T_2$, then the partition $\{\{00, 01\}, \{10, 11\}\}$ of $\mathcal{L}_2$ is necessarily preserved by G .



Definition

The action of G on T is **tree-primitive** if for every n , the only partitions of $\mathcal{L}_n$ preserved by the action of G are the partition that are already preserved by $\text{Aut}(T)$.

Trivial branch kernel

Definition

A branch group $G \leq \text{Aut}(T)$ is said to have **trivial branch kernel** if for every normal subgroup $N \trianglelefteq G$ of finite index, there exists n such that $\text{Rist}_G(n) \leq N$.

A branch group G can be endowed with the profinite topology (generated by the finite index normal subgroups and their subgroups), but also with the branch topology (generated by the $\text{Rist}_G(n)$ and their cosets). G has trivial branch kernel if and only if these two topologies coincide. Branch group with the congruence subgroup property have trivial branch kernel.

The subgroup induction property

Definition

Let $G \leq \text{Aut}(T)$ be a self-similar group. A family $\mathcal{C}$ of subgroups of G is said to be **inductive** if

1. Both $\{1\}$ and G belong to $\mathcal{C}$,
2. If $H \in \mathcal{C}$ and L contains H as a finite index subgroup, then $L \in \mathcal{C}$,
3. If H is a finitely generated subgroup of $\text{Stab}_G(\mathcal{L}_1)$ and all first level sections of H are in $\mathcal{C}$, then H is in $\mathcal{C}$.

It is clear that the collection $\mathcal{C}$ of finitely generated subgroups of G is inductive.

Definition (Grigorchuk-Wilson, 2003)

A self-similar group G has the **subgroup induction property** (SIP for short) if for any inductive class of subgroups $\mathcal{C}$, each finitely generated subgroup of G is contained in $\mathcal{C}$.

The subgroup induction property: consequences

Theorem (Francoeur-L., 2025, (Grigorchuk-Wilson 2003, Garrido 2016))

Let G be a finitely generated branch group with the subgroup induction property. Then, G is torsion and just infinite.

1. *If $G \leq \text{Aut}(T_d)$ is self-replicating and $H \leq G$ is finitely generated, then H is commensurable with one of $1, G, G^2, \dots, G^{d-1}$;*
2. *If $G \leq \text{Aut}(T_p)$ with p prime, then every maximal subgroup of G is of finite index;*
3. *If G is self-similar, then it is LERF;*
4. *If $G \leq \text{Aut}(T_p)$ with p prime is self-similar and $H \leq G$ is finitely generated, then every maximal subgroup of H is of finite index in H and every weakly maximal subgroup of H is closed in the profinite topology.*

The subgroup induction property: examples

The following groups have the subgroup induction property:

- ▶ The first Grigorchuk group [Grigorchuk-Wilson, 2003];
- ▶ The Gupta-Sidki 3-group [Garrido, 2016];
- ▶ Torsion GGS groups (for a prime p) [Francoeur-L., 2025].

Consequences of the main result

Let G be either the first Grigorchuk group or torsion GGS group (for some prime p). Then;

- ▶ G is LERF (alternative proof);
- ▶ We have a classification of weakly maximal subgroups of G [L. 2025];
- ▶ We have a classification of finitely generated self-commensurating subgroups of G (can be used to construct irreducible representations of G) [Francoeur-L.-Nagnibeda 2025⁺⁺].

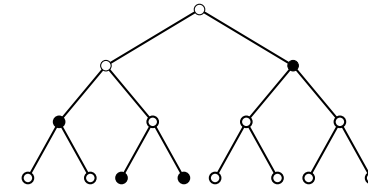
Some projects

Use Theorem A to:

- ▶ show that $\mathfrak{G}$ has the Ribes-Zalesskiĭ property;
- ▶ have a algorithm that given two finitely generated subgroups H and K of $\mathfrak{G}$ decide whenever $H \cap K$ is finitely generated or not.

A few words on the proof

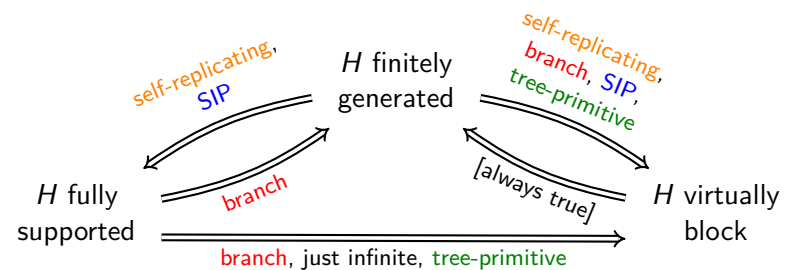
A subgroup $H \leq G \leq \text{Aut}(T)$ is **fully supported** if there exists a transversal $V \subseteq T$ such that for all $v \in V$, $\pi_v(\text{Stab}_H(v))$ is finite or $= G$.



A few words on the proof

A subgroup $H \leq G \leq \text{Aut}(T)$ is **fully supported** if there exists a transversal $V \subseteq T$ such that for all $v \in V$, $\pi_v(\text{Stab}_H(v))$ is finite or $= G$.

Let $G \leq \text{Aut}(X^*)$ be a finitely generated group and $H \leq G$ a subgroup. Then we have the following implications:



Virtually block implies finitely generated

Since G finitely generated, so is K as well as any block subgroup. Indeed, blocks subgroups are isomorphic to K^n for some n . Therefore, all virtually block subgroups are finitely generated.

Finitely generated implies fully supported

Let $G \leq \text{Aut}(T)$ be a self-replicating subgroup. One easily show that the class $\mathcal{C}$ of finitely generated fully supported subgroups of G is inductive. By the subgroup induction property, we conclude that $\mathcal{C}$ is the whole class of finitely generated subgroups.

Fully supported implies virtually block

Let H be a fully supported subgroup and let V be the transversal witnessing it. Let $F \subseteq V$ be the subset of vertices such that $\pi_v(\text{Stab}_H(v)) = G$. For any $v \in F$, one can find a *minimal dependance set* $W \subseteq F$ such that

$$\pi_v \left(\text{Stab}(W) \cap \prod_{w \in W} \text{Rist}_H(w) \right) \neq \{1\}.$$

One can show that: W does not depend on v . Moreover, if G is branch, just infinite and with a tree primitive action, and $W = \{w_1, \dots, w_n\}$ is a minimal dependance set, then there exists $Z = \{z_1 \leq w_1, \dots, z_n \leq w_n\}$ such that $\text{Stab}(Z) \cap \prod_{z \in Z} \text{Rist}_H(z)$ is a diagonal subgroup.

Some elements of the proof

A throughout study of **almost normal subgroups**: $H \leq G$ such that H is normal in a finite index subgroup of G .

Definition

Let $H \leq G$ be an almost normal subgroup. An almost normal subgroup $K \leq G$ is a **complement** for H if: $H \cap K = \{1\}$, $[H, K] = \{1\}$ (i.e. HK is the direct product of H and K) and HK has finite index in G .

Proposition

Let G be a just infinite branch group and let $H \leq G$ be an almost normal subgroup. Then H admits an almost normal complement.

More groups satisfying the hypothesis of Theorem A?

Want G finitely generated, self-replicating, regularly branch, with trivial branch kernel, a tree-primitive action and the subgroup induction property.

Many examples of finitely generated, self-replicating, regularly branch groups with the congruence subgroup property (and hence with trivial branch kernel) are known.

It remains to show that some of these examples have a tree-primitive action and the subgroup induction property.

Checking for tree-primitive actions

Tree-primitivity is about the action on the whole tree. Hopefully, it is enough to check it on the first two levels only.

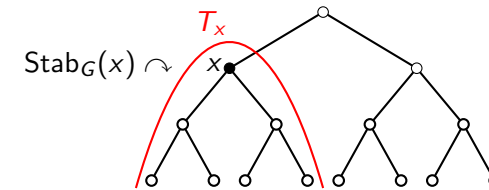
Theorem

Let G be a self-replicating group acting spherically transitively on T such that the action of G on the first level $\mathcal{L}_1$ is primitive. Suppose moreover that:

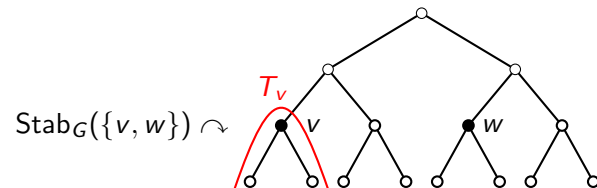
1. $\text{Stab}_G(\mathcal{L}_1)$ acts spherically transitively on T_x for all $x \in \mathcal{L}_1$;
2. For every $v, w \in \mathcal{L}_2$, the subgroup $\text{Stab}_G(\{v, w\})$ acts spherically transitively on T_v ;
3. There exists $x_0 \neq y_0 \in \mathcal{L}_1$ such that for every $v \in T_{x_0} \cap \mathcal{L}_2$ and $w \in T_{y_0} \cap \mathcal{L}_2$, we have $\text{Stab}_G(v) \cap \text{Stab}_G(y_0) \not\leq \text{Stab}_G(w)$.

Then the action of G on X is tree-primitive.

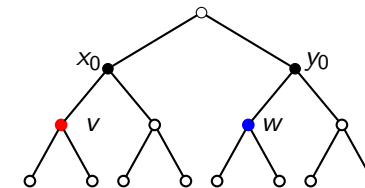
1. $\text{Stab}_G(\mathcal{L}_1)$ acts spherically transitively on T_x for all $x \in \mathcal{L}_1$;



2. For every $v, w \in \mathcal{L}_2$, the subgroup $\text{Stab}_G(\{v, w\})$ acts spherically transitively on T_v ;



There exists $x_0 \neq y_0 \in \mathcal{L}_1$ such that for every $v \in T_{x_0} \cap \mathcal{L}_2$ and $w \in T_{y_0} \cap \mathcal{L}_2$, we have $\text{Stab}_G(v) \cap \text{Stab}_G(y_0) \not\leq \text{Stab}_G(w)$.



There exists g fixing y_0 and v , but moving w .

Checking for the subgroup induction property

Definition

Let $G \leq \text{Aut}(T)$ be a self-similar group. A family $\mathcal{C}$ of subgroups of G is said to be **strongly inductive** if

1. Both $\{1\}$ and G belong to $\mathcal{C}$,
2. If L contains H as a finite index subgroup, then $L \in \mathcal{C} \iff H \in \mathcal{C}$,
3. If H is a finitely generated subgroup of $\text{Stab}_G(\mathcal{L}_1)$ and all first level sections of H are in $\mathcal{C}$, then H is in $\mathcal{C}$.

Definition

A self-similar group G has the **weak subgroup induction property** (wSIP for short) if for any strongly inductive class of subgroups $\mathcal{C}$, each finitely generated subgroup of G is contained in $\mathcal{C}$.

It is clear that $\text{SIP} \implies \text{wSIP}$.

The weak subgroup induction property

Proposition (Francoeur-L. 2025)

Let G be a self-replicating group with the congruence subgroup property and such that for every $v \in \mathcal{L}_n$, $\pi_v(\text{Stab}_G(\mathcal{L}_n)) = G$. Then if G has wSIP it also has SIP.

Proposition (Francoeur-L. 2025)

Let G be a finitely generated self-similar group with SIP. Then for every $v \in \mathcal{L}_1$, $\pi_v(\text{Stab}_G(\mathcal{L}_n)) = G$.

Proposition (Grigorchuk-L.-Nagnibeda 2021)

Let G be a self-replicating group. Then G has the wSIP if and only if: for every finitely generated $H \leq G$, there is a transversal V of T such that for every $v \in V$ the section $\pi_v(\text{Stab}_H(V))$ is either trivial or of finite index in G .

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